

DECISION SUPPORT SYSTEM (DSS)

EVALUATION BASED ON DISTANCE FROM AVERAGE SOLUTION - EDAS

TEACHING TEAM

DECISION SUPPORT SYSTEM COURSE YEAR 24/25

Evaluation Based on Distance from Average Solution (EDAS)



This method was introduced by Keshavarz Ghorabae in 2015 to deal with MCDM problems



The EDAS method uses an average solution for alternative assessment by calculating the positive mean distance (PDA) and the negative mean distance (NDA)



This method is particularly useful when conflicting criteria must be considered (benefit and cost), stable when various weighting criteria are used, and consistent with other methods

Evaluation Based on Distance from Average Solution (EDAS)

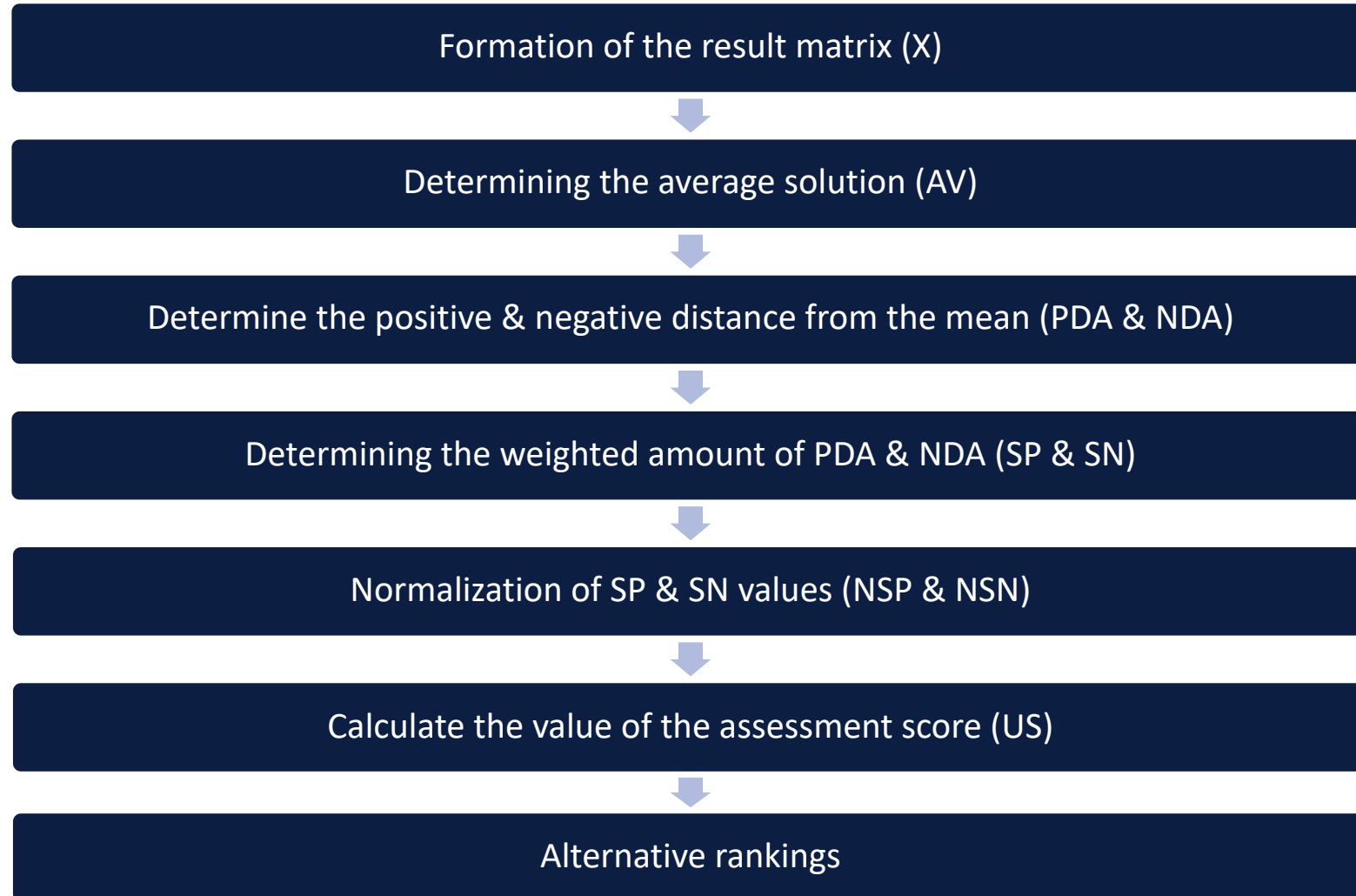


Multicriteria decision-making final results based on the highest Appraisal Score (AS) score to get the best choice of all alternatives



Simplicity and faster calculations are advantages of this method, especially since these advantages do not affect the accuracy of calculations

Stages of the EDAS Method



1. Formation of the Decision Matrix (X)

In the decision matrix (X), the row shows the alternative and the column shows the criteria. The decision matrix shows the performance of each alternative against various criteria.

$$X = \begin{bmatrix} x_{11} & \dots & x_{12} & \dots & x_{1n} \\ x_{21} & \dots & x_{22} & \dots & x_{2n} \\ x_{m1} & \dots & x_{m2} & \dots & x_{mn} \end{bmatrix}$$

Information:

m = alternative

N = criteria

2. Determining the Average Solution (AV)

The determination of the average solution (AV) according to the criteria specified using the following equations:

$$AV_j = \frac{\sum_{i=1}^m X_{ij}}{m}$$

Information:

AV_j = average solution

X_{ij} = the value of the criteria of the alternative

M = alternative

3. Determining the Positive/Negative Distance from the Average (PDA/NDA)

Calculate the positive distance from the mean matrix (PDA) and the negative distance from the mean matrix (NDA) according to the type of criteria (benefit and cost) using the equation:

$$\begin{aligned} PDA_{ij} &= \left\{ \frac{\max(0, (x_{ij} - AV_j))}{AV_j} \right\} \text{ kriteria benefit} \\ PDA_{ij} &= \left\{ \frac{\max(0, (AV_j - x_{ij}))}{AV_j} \right\} \text{ kriteria cost} \end{aligned}$$
$$\begin{aligned} NDA_{ij} &= \left\{ \frac{\max(0, (AV_j - x_{ij}))}{AV_j} \right\} \text{ kriteria benefit} \\ NDA_{ij} &= \left\{ \frac{\max(0, (x_{ij} - AV_j))}{AV_j} \right\} \text{ kriteria cost} \end{aligned}$$

4. Determining the Weighted Amount of PDA/NDA (SP/SN)

Determine the weighted amount of PDA and NDA for all alternatives with the following equation:

$$SP_i = \sum_{j=1}^n w_j \times PDA_{ij}$$
$$SN_i = \sum_{j=1}^n w_j \times NDA_{ij}$$

The values of SP_i and SN_i , respectively, are the weighted sum values of PDA and NDA for each of the i -th alternatives.

5. Normalization of SP/SN VALUE (NSP/NSN)

The next stage is to calculate the normalization values of SP and SN for all alternatives with the following equation:

$$NSP_i = \frac{SP_i}{\max(SP_i)}$$
$$NSN_i = 1 - \frac{SN_i}{\max(SN_i)}$$

6. Calculating the Value of the Assessment Score (AS)

Calculate the Apraisal Score of the assessment with the following equation:

$$AS_i = \frac{1}{2} (NSP_i + NSN_i)$$

7. Alternative Ranking

Ranking from the Assessment Score (AS) from the highest to the lowest. The alternative with the highest value indicates the best alternative.

Case Study 1 – Worker Recruitment

- A company will recruit 5 prospective workers for the position of machine operator.
- Only 2 positions are needed with the following criteria:

C1	Work experience	Benefit
C2	Education	Benefit
C3	Age	Benefit
C4	Marital status	Cost
C5	Address	Cost

Weighting (w)

Kriteria	Bobot
C1	0,3
C2	0,2
C3	0,2
C4	0,15
C5	0,15
TOTAL	1

- ✓ Ada 5 orang yang menjadi kandidat (alternatif) yaitu:
- ✓ Doni Prakosa (A1)
- ✓ Dion Pratama (A2)
- ✓ Dina Ayu Palupi (A3)
- ✓ Dini Ambarwati (A4)
- ✓ Danu Nugraha (A5)

Alternative Assessment for Each Criterion

Alternatif	Kriteria				
	C1	C2	C3	C4	C5
A1	0,5	1	0,7	0,7	0,8
A2	0,8	0,7	1	0,5	1
A3	1	0,3	0,4	0,7	1
A4	0,2	1	0,5	0,9	0,7
A5	1	0,7	0,4	0,7	1

1. Formation of the Decision Matrix (X)

$$X = \begin{bmatrix} 0,5 & 1 & 0,7 & 0,7 & 0,8 \\ 0,8 & 0,7 & 1 & 0,5 & 1 \\ 1 & 0,3 & 0,4 & 0,7 & 1 \\ 0,2 & 1 & 0,5 & 0,9 & 0,7 \\ 1 & 0,7 & 0,4 & 0,7 & 1 \end{bmatrix}$$

Alternatif	Kriteria				
	C1	C2	C3	C4	C5
A1	0,5	1	0,7	0,7	0,8
A2	0,8	0,7	1	0,5	1
A3	1	0,3	0,4	0,7	1
A4	0,2	1	0,5	0,9	0,7
A5	1	0,7	0,4	0,7	1

2. Determining the AV Average Solution

$$AV_1 = \frac{0,5+0,8+1+0,2+1}{5} = \frac{3,5}{5} = 0,7$$

$$AV_2 = \frac{1+0,7+0,3+1+0,7}{5} = \frac{3,7}{5} = 0,74$$

$$AV_3 = \frac{0,7+1+0,4+0,5+0,4}{5} = \frac{3}{5} = 0,6$$

$$AV_4 = \frac{0,7+0,5+0,7+0,9+0,7}{5} = \frac{3,5}{5} = 0,7$$

$$AV_5 = \frac{0,8+1+1+0,7+1}{5} = \frac{4,5}{5} = 0,9$$

$$AV = [0,7 \quad 0,74 \quad 0,6 \quad 0,7 \quad 0,9]$$

$$X = \begin{bmatrix} 0,5 & 1 & 0,7 & 0,7 & 0,8 \\ 0,8 & 0,7 & 1 & 0,5 & 1 \\ 1 & 0,3 & 0,4 & 0,7 & 1 \\ 0,2 & 1 & 0,5 & 0,9 & 0,7 \\ 1 & 0,7 & 0,4 & 0,7 & 1 \end{bmatrix}$$

$$AV_j = \frac{\sum_{i=1}^m X_{ij}}{m}$$

3.1 Determining Positive Distance from Mean (PDA)

Kriteria C1 :

$$PDA_{11} = \max\left(0, \frac{(0,5-0,7)}{0,7}\right) = \left(0, \frac{-0,2}{0,7}\right) = (0, -0,286) = 0$$

$$PDA_{21} = \max\left(0, \frac{(0,8-0,7)}{0,7}\right) = \left(0, \frac{0,1}{0,7}\right) = (0, 0,143) = 0,143$$

$$PDA_{31} = \max\left(0, \frac{(1-0,7)}{0,7}\right) = \left(0, \frac{0,3}{0,7}\right) = (0, 0,429) = 0,429$$

$$PDA_{41} = \max\left(0, \frac{(0,2-0,7)}{0,7}\right) = \left(0, \frac{-0,5}{0,7}\right) = (0, -0,714) = 0$$

$$PDA_{51} = \max\left(0, \frac{(1-0,7)}{0,7}\right) = \left(0, \frac{0,3}{0,7}\right) = (0, 0,429) = 0,429$$

$$PDA_{ij} = \left\{ \frac{\max(0, (x_{ij} - AV_j))}{AV_j} \right\} \text{ kriteria benefi}$$

$$PDA_{ij} = \left\{ \frac{\max(0, (AV_j - x_{ij}))}{AV_i} \right\} \text{ kriteria cost}$$

$$X = \begin{array}{ccccc} 0,5 & 1 & 0,7 & 0,7 & 0,8 \\ 0,8 & 0,7 & 1 & 0,5 & 1 \\ 1 & 0,3 & 0,4 & 0,7 & 1 \\ 0,2 & 1 & 0,5 & 0,9 & 0,7 \\ 1 & 0,7 & 0,4 & 0,7 & 1 \end{array}$$

$$AV = [0,7 \ 0,74 \ 0,6 \ 0,7 \ 0,9]$$

Positive Distance Data from Average Solution (PDA)

$$PDA_{ij} = \begin{bmatrix} 0 & 0,351 & 0,167 & 0 & 0,111 \\ 0,143 & 0 & 0,667 & 0,286 & 0 \\ 0,429 & 0 & 0 & 0 & 0 \\ 0 & 0,351 & 0 & 0 & 0,222 \\ 0,429 & 0 & 0 & 0 & 0 \end{bmatrix}$$

3.2 Determining the Negative Distance from the Mean (NDA)

Kriteria C1 :

$$NDA_{11} = \max\left(0, \frac{(0,7-0,5)}{0,7}\right) = \left(0, \frac{0,2}{0,7}\right) = (0, 0,286) = 0,286$$

$$NDA_{21} = \max\left(0, \frac{(0,7-0,8)}{0,7}\right) = \left(0, \frac{-0,1}{0,7}\right) = (0, -0,143) = 0$$

$$NDA_{31} = \max\left(0, \frac{(0,7-1)}{0,7}\right) = \left(0, \frac{-0,3}{0,7}\right) = (0, -0,429) = 0$$

$$NDA_{41} = \max\left(0, \frac{(0,7-0,2)}{0,7}\right) = \left(0, \frac{0,5}{0,7}\right) = (0, 0,714) = 0,714$$

$$NDA_{51} = \max\left(0, \frac{(0,7-1)}{0,7}\right) = \left(0, \frac{-0,3}{0,7}\right) = (0, -0,429) = 0$$

$$VDA_{ij} = \left\{ \frac{\max(0, (AV_j - x_{ij}))}{AV_j} \right\} \text{ kriteria benefit}$$

$$NDA_{ij} = \left\{ \frac{\max(0, (x_{ij} - AV_j))}{AV_i} \right\} \text{ kriteria cost}$$

$$X = \begin{array}{ccccc} 0,5 & 1 & 0,7 & 0,7 & 0,8 \\ 0,8 & 0,7 & 1 & 0,5 & 1 \\ 1 & 0,3 & 0,4 & 0,7 & 1 \\ 0,2 & 1 & 0,5 & 0,9 & 0,7 \\ 1 & 0,7 & 0,4 & 0,7 & 1 \end{array}$$

$$AV = [0,7 \ 0,74 \ 0,6 \ 0,7 \ 0,9]$$

Negative Distance Data from Average Solution (NDA)



$$NDA_{ij} = \begin{bmatrix} 0,286 & 0 & 0 & 0 & 0 \\ 0 & 0,054 & 0 & 0 & 0,111 \\ 0 & 0,595 & 0,333 & 0 & 0,111 \\ 0,714 & 0 & 0,167 & 0,286 & 0 \\ 0 & 0,054 & 0,333 & 0 & 0,111 \end{bmatrix}$$

4. Determining the Weighted Amount of PDA/NDA (SP/SN)



$$SP_i = \sum_{j=1}^n w_j \times PDA_{ij}$$

$$SN_i = \sum_{j=1}^n w_j \times NDA_{ij}$$

$$SP_1 = (0,3*0)+(0,2*0,351)+(0,2*0,167)+(0,15*0)+(0,15*0,111) = 0,120$$

$$SP_2 = (0,3*0,143)+(0,2*0)+(0,2*0,667)+(0,15*0,286)+(0,15*0) = 0,219$$

$$SP_3 = (0,3*0,429)+(0,2*0)+(0,2*0)+(0,15*0)+(0,15*0) = 0,129$$

$$SP_4 = (0,3*0)+(0,2*0,351)+(0,2*0)+(0,15*0)+(0,15*0,222) = 0,104$$

$$SP_5 = (0,3*0,429)+(0,2*0)+(0,2*0)+(0,15*0)+(0,15*0) = 0,129$$

$$SN_1 = (0,3*0,286)+(0,2*0)+(0,2*0)+(0,15*0)+(0,15*0) = 0,086$$

$$SN_2 = (0,3*0)+(0,2*0,054)+(0,2*0)+(0,15*0)+(0,15*0,111) = 0,027$$

$$SN_3 = (0,3*0)+(0,2*0,595)+(0,2*0,333)+(0,15*0)+(0,15*0,111) = 0,202$$

$$SN_4 = (0,3*0,714)+(0,2*0)+(0,2*0,167)+(0,15*0,286)+(0,15*0) = 0,290$$

$$SN_5 = (0,3*0)+(0,2*0,054)+(0,2*0,333)+(0,15*0)+(0,15*0,111) = 0,094$$

PDA_{ij}	0	0,351	0,167	0	0,111
	0,143	0	0,667	0,286	0
	0,429	0	0	0	0
	0	0,351	0	0	0,222
	0,429	0	0	0	0

NDA_{ij}	0,286	0	0	0	0
	0	0,054	0	0	0,111
	0	0,595	0,333	0	0,111
	0,714	0	0,167	0,286	0
	0	0,054	0,333	0	0,111

Kriteria	Bobot
C1	0,3
C2	0,2
C3	0,2
C4	0,15
C5	0,15
TOTAL	1

5. Normalization of SP/SN VALUE (NSP/NSN)

$$NSP_1 = \frac{0,120}{0,219} = 0,549$$

$$NSN_1 = 1 - \frac{0,086}{0,290} = 0,705$$

$$NSP_2 = \frac{0,219}{0,219} = 1$$

$$NSN_2 = 1 - \frac{0,027}{0,290} = 0,905$$

$$NSP_3 = \frac{0,129}{0,219} = 0,587$$

$$NSN_3 = 1 - \frac{0,202}{0,290} = 0,304$$

$$NSP_4 = \frac{0,104}{0,219} = 0,473$$

$$NSN_4 = 1 - \frac{0,290}{0,290} = 0$$

$$NSP_5 = \frac{0,129}{0,219} = 0,587$$

$$NSN_5 = 1 - \frac{0,094}{0,290} = 0,676$$

$$NSP_i = \frac{SP_i}{\max(SP_i)}$$

$$NSN_i = 1 - \frac{SN_i}{\max(SN_i)}$$

$$SP_1 = 0,120$$

$$SN_1 = 0,086$$

$$SP_2 = 0,219$$

$$SN_2 = 0,027$$

$$SP_3 = 0,129$$

$$SN_3 = 0,202$$

$$SP_4 = 0,104$$

$$SN_4 = 0,290$$

$$SP_5 = 0,129$$

$$SN_5 = 0,094$$

6. Calculating the Value of the Assessment Score(AS)

$$AS_1 = \frac{1}{2} (0,549 + 0,705) = 0,627$$

$$AS_2 = \frac{1}{2} (1 + 0,905) = 0,953$$

$$AS_3 = \frac{1}{2} (0,587 + 0,304) = 0,445$$

$$AS_4 = \frac{1}{2} (0,473 + 0) = 0,236$$

$$AS_5 = \frac{1}{2} (0,587 + 0,676) = 0,631$$

$$AS_i = \frac{1}{2} (NSP_i + NSN_i)$$

$$NSP_1 = 0,549$$

$$NSN_1 = 0,705$$

$$NSP_2 = 1$$

$$NSN_2 = 0,905$$

$$NSP_3 = 0,587$$

$$NSN_3 = 0,304$$

$$NSP_4 = 0,473$$

$$NSN_4 = 0$$

$$NSP_5 = 0,587$$

$$NSN_5 = 0,676$$

7. Alternative Ranking

$$AS_1 = 0,627$$

$$AS_2 = 0,953$$

$$AS_3 = 0,445$$

$$AS_4 = 0,236$$

$$AS_5 = 0,631$$

- The largest value is $A_2 = 0.953$ and $A_5 = 0.631$ so that **Dion Pratama** and **Danu Nugraha** are the alternatives chosen as the best alternative for the position of machine operator.

